GSE 518
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Suggested Solutions to Problem Set 10

1. From Theorem 5.2.6 we know that $E(s^2) = \sigma_0^2$ regardless of the distribution of X_i. Therefore, s^2 is an unbiased estimator of σ_0^2. Moreover, in this problem $X_1, X_2, \ldots, X_n$ is a random sample from a normal distribution $N(\mu_0, \sigma_0^2)$; therefore, by question 4 in problem set 7 we know $Var(s^2) = 2\sigma_0^4/(n-1)$. Hence, we have

$$\lim_{n\to\infty} Bias\left(s^2\right) = 0$$

and

$$\lim_{n\to\infty} Var\left(s^2\right) = \lim_{n\to\infty} \frac{2\sigma_0^4}{n-1} = 0$$

Therefore, s^2 converges in mean square error to σ_0^2, which implies s^2 converges in probability to σ_0^2, i.e., $s^2 \xrightarrow{P} \sigma_0^2$. Hence, s^2 is consistent for σ_0^2.

2. First we obtain the method of moments estimator of θ_0. Since we have only one parameter, we need only one moment. The first moment of X in this case is

$$\begin{aligned} E(X) &= \int_{-1}^{1} x\frac{1}{2}(1+\theta_0 x)\,dx \\ &= \frac{1}{2}\int_{-1}^{1}\left(x+\theta_0 x^2\right)dx \\ &= \frac{1}{2}\left[\frac{x^2}{2}+\theta_0\frac{x^3}{3}\right]\Bigg|_{-1}^{1} \\ &= \frac{1}{2}\left[\frac{1}{2}+\frac{\theta_0}{3}-\frac{1}{2}+\frac{\theta_0}{3}\right] \\ &= \frac{\theta}{3} \end{aligned}$$

Then, to find the MOM estimator we set $E(X) = \frac{\theta}{3} = \overline{X}$. Hence, $\widehat{\theta}_{mom} = 3\overline{X}$ (where as usual $\overline{X}$ is the sample average).
We now need to show that the estimator $\widehat{\theta}_{mom}$ is consistent. As in question 1, we need the mean and variance of $\widehat{\theta}_{mom}$. Its mean is given by $E(\widehat{\theta}_{mom}) = E\left(3\overline{X}\right) = 3E\left(\overline{X}\right) = 3\frac{\theta_0}{3} = \theta_0$ (since we know from Theorem 5.2.6 that $E\left(\overline{X}\right) = E(X_i)$. Or you could also show this by following the same steps as in the proof of that theorem). Hence we have

$$\lim_{n\to\infty} Bias\left(\widehat{\theta}_{mom}\right) = 0$$

since $\widehat{\theta}_{mom}$ is unbiased for θ_0.

We now need the variance of $\widehat{\theta}_{mom}$. To obtain the variance of $\widehat{\theta}_{mom}$ we need the variance of X_i. Using the pdf given in the problem we can calculate

$$\begin{aligned} E\left(X^2\right) &= \int_{-1}^{1} x^2 \frac{1}{2}\left(1+\theta_0 x\right) dx \\ &= \frac{1}{2}\int_{-1}^{1}\left(x^2+\theta_0 x^3\right) dx \\ &= \frac{1}{2}\left[\frac{x^3}{3}+\theta_0\frac{x^4}{4}\right]\Big|_{-1}^{1} \\ &= \frac{1}{2}\left[\frac{1}{3}+\frac{\theta_0}{4}+\frac{1}{3}-\frac{\theta_0}{4}\right] \\ &= \frac{1}{3} \end{aligned}$$

Hence, we have $Var\left(X_i\right) = E\left(X_i^2\right) - \left[E\left(X_i\right)\right]^2 = \frac{1}{3} - \frac{\theta_0^2}{9}$. Then, the variance of $\widehat{\theta}_{mom}$ is given by

$$\begin{aligned} Var\left(\widehat{\theta}_{mom}\right) &= Var\left(3\overline{X}\right) \\ &= 9Var\left(\overline{X}\right) \\ &= 9\frac{Var\left(X_i\right)}{n} \\ &= \frac{9}{n}\left[\frac{1}{3}-\frac{\theta_0^2}{9}\right] \\ &= \frac{1}{n}\left[3-\theta_0^2\right] \end{aligned}$$

where in the third line we have used the result from Theorem 5.2.6 that $Var\left(\overline{X}\right) = Var\left(X_i\right)/n$ (or you could also show this by following the same steps as in the proof of that theorem). Therefore, we have

$$\lim_{n\to\infty} Var\left(\widehat{\theta}_{mom}\right) = \lim_{n\to\infty}\frac{1}{n}\left[3-\theta_0^2\right] = 0$$

Thus, since $\lim_{n\to\infty} Bias\left(\widehat{\theta}_{mom}\right) = 0$ and $\lim_{n\to\infty} Var\left(\widehat{\theta}_{mom}\right) = 0$, we have that $\widehat{\theta}_{mom} \xrightarrow{P} \theta_0$ and $\widehat{\theta}_{mom}$ is consistent for θ.

3.

(a) We know that $\widehat{\sigma}^2_{MLE}$ is the maximum likelihood estimator of σ_0^2, and $\sigma_0 = \left(\sigma_0^2\right)^{1/2}$. Therefore, using the "Invariance Property of the MLE" (see classnotes or Theorem 7.2.10 in CB) we have that $\widehat{\sigma}_{MLE} = \left(\widehat{\sigma}^2_{MLE}\right)^{1/2}$ is the MLE for σ_0.

(b) Note that in this case we know the limiting distribution of $\sqrt{n}(\widehat{\sigma}^2_{MLE} - \sigma_0^2)$ (it is given in the problem), and we want to derive the asymptotic distribution of a *function* of $\widehat{\sigma}^2_{MLE}$,

which is $\widehat{\sigma}_{MLE} = (\widehat{\sigma}^2_{MLE})^{1/2}$. Therefore, we use the Delta Method (see classnotes or Theorem 5.5.24). In this case, we have that $\sigma_0 = (\sigma_0^2)^{1/2}$, so

$$\frac{d\left(\sigma_0^2\right)^{1/2}}{d\sigma_0^2} = \frac{1}{2}\left(\sigma_0^2\right)^{-1/2}$$

Applying the Delta Method we have

$$\sqrt{n}(\widehat{\sigma}_{MLE} - \sigma_0) \xrightarrow{d} N\left(0, \left[\frac{1}{2}\left(\sigma_0^2\right)^{-1/2}\right]^2 2\sigma_0^4\right),$$

or simply

$$\sqrt{n}(\widehat{\sigma}_{MLE} - \sigma_0) \xrightarrow{d} N(0, \frac{1}{2}\sigma_0^2).$$

Hence, the asymptotic (or approximate) distribution of $\widehat{\sigma}_{MLE}$ is:

$$\widehat{\sigma}_{MLE} \overset{A}{\sim} N(\sigma_0, \frac{1}{2n}\sigma_0^2)$$

4.

(a) From the theorem seen in class regarding the asymptotic properties of maximum likelihood estimators (Theorem 10.1.12), we know that the MLE is consistent, asymptotically normal and asymptotically efficient. Hence, in this part we derive the MLE for θ_0. Note that you did this in problem set 8 question 3(a), and you found

$$\widehat{\theta}_{MLE} = \frac{-n}{\sum_{i=1}^n \ln(X_i)}$$

(b) From the same theorem mentioned in part (a), we know that

$$\sqrt{n}(\widehat{\theta}_{MLE} - \theta_0) \xrightarrow{d} N\left(0, \left[I(\theta_0)\right]^{-1}\right) \tag{1}$$

where $I(\theta_0)$ is the Fisher Information Matrix and equals

$$I(\theta_0) = E\left(\left[S(\theta_0, X)\right]^2\right) = Var\left(S(\theta_0, X)\right) = -E\left(\frac{\partial S(\theta_0, X)}{\partial \theta}\right) \tag{2}$$

where $S(\theta_0, X)$ is the score function $S(\theta_0, X) = \frac{\partial \ln f(\theta_0, x)}{\partial \theta}$.
In this problem, we have that $\ln f(x|\theta) = \ln(\theta) + (\theta - 1)\ln(x)$, so the score function is

$$S(\theta_0, X) = \frac{\partial \ln f(\theta_0, x)}{\partial \theta} = \frac{1}{\theta_0} + \ln(X)$$

In this case the easier formula to use to get the information matrix is the last one in equation (2) (otherwise you need to get the variance of $\ln(X)$, or the expected value of $\ln(X)$ and $[\ln(X)]^2$). Using that formula we find

$$I(\theta_0) = -E\left(\frac{\partial S(\theta_0, X)}{\partial \theta}\right) = -E\left[-\frac{1}{\theta_0^2}\right] = \frac{1}{\theta_0^2}$$

Using (1) we find that the limiting distribution of $\sqrt{n}(\widehat{\theta}_{MLE} - \theta_0)$ is

$$\sqrt{n}(\widehat{\theta}_{MLE} - \theta_0) \xrightarrow{d} N\left(0, \theta_0^2\right) \tag{3}$$

Finally, using our result in (3), the asymptotic (or approximate) distribution of our estimator $\widehat{\theta}_{MLE}$ is:

$$\widehat{\theta}_{MLE} \overset{A}{\sim} N\left(\theta_0, \theta_0^2/n\right).$$

(c) From part (b), we have that for large n we can approximate the distribution of $\widehat{\theta}_{MLE}$ as $N\left(\theta_0, \theta_0^2/n\right)$. Thus, the approximate (or asymptotic) variance of $\widehat{\theta}_{MLE}$ is θ_0^2/n, or for short hand, write $AVar(\widehat{\theta}_{MLE}) = \theta_0^2/n$. Now, how can we estimate $AVar(\widehat{\theta}_{MLE})$? Note that $AVar(\widehat{\theta}_{MLE})$ is a function of θ_0, and we already have the MLE for θ_0. Therefore, using the Invariance Property of the MLE (see question 3a), we have that we can estimate the asymptotic variance of $\widehat{\theta}_{MLE}$ as

$$\widehat{AVar}(\widehat{\theta}_{MLE}) = \frac{\widehat{\theta}_{MLE}^2}{n} = \frac{n}{\left[\sum_{i=1}^n \ln\left(X_i\right)\right]^2}$$

5. In this case the log of the pdf is given by

$$\ln f\left(x|\theta\right) = \ln\left(\theta\right) - \theta x$$

and therefore the log-likelihood function is given by

$$L\left(\theta|X_1, ..., X_n\right) = n\ln\left(\theta\right) - \theta\sum_{i=1}^n X_i$$

Maximizing this function with respect to θ we find

$$\frac{\partial L\left(\theta|X_1, ..., X_n\right)}{\partial\theta} = \frac{n}{\theta} - \sum_{i=1}^n X_i = 0$$

Solving for θ we find the MLE for θ_0 is

$$\widehat{\theta}_{MLE} = \frac{1}{\overline{X}} \tag{4}$$

We know the limiting distribution of the ML estimator is given by (1). Therefore, we only need to derive the Fisher Information matrix using any of the formulas in (2). In this problem we have that $\ln f\left(x|\theta\right) = \ln\left(\theta\right) - \theta X$, so the score function is

$$S\left(\theta_0, X\right) = \frac{\partial \ln f\left(\theta_0, x\right)}{\partial\theta} = \frac{1}{\theta_0} - X$$

In this case probably the easier formula to use to get the information matrix is the last one in equation (2) (otherwise you need to get the variance of X or the expected value of X and X^2). Using that formula we find

$$I(\theta_0) = -E\left(\frac{\partial S(\theta_0, X)}{\partial \theta}\right) = -E\left[-\frac{1}{\theta_0^2}\right] = \frac{1}{\theta_0^2}$$

Thus, using (1) we find that

$$\sqrt{n}(\widehat{\theta}_{MLE} - \theta_0) \xrightarrow{d} N\left(0, \theta_0^2\right),$$

so the asymptotic distribution of $\widehat{\theta}_{MLE}$ is:

$$\widehat{\theta}_{MLE} \overset{A}{\sim} N\left(\theta_0, \theta_0^2/n\right)$$

6.

(a) In question 2(a) in problem set 8 you derived the MLE for θ_0, which is given by

$$\widehat{\theta}_{MLE} = \frac{1}{n}\sum_{i=1}^{n} X_i = \overline{X}$$

Moreover, in question 2(a) from problem set 9 you derived the Cramèr-Rao Lower Bound for any unbiased estimator of the parameter of a Poisson distribution. There you found that the Fisher Information matrix was given by $I(\theta_0) = 1/\theta_0$. Therefore, using equation (1) we have that the limiting distribution of the MLE for θ_0 in this problem is given by

$$\sqrt{n}(\widehat{\theta}_{MLE} - \theta_0) \xrightarrow{d} N(0, \theta_0) \tag{5}$$

(b) This question is very similar to 3. Note that we know that $\widehat{\theta}_{MLE} = \overline{X}$ is the maximum likelihood estimator of θ_0, and we want to estimate $\gamma_0 = \Pr(X = 0) = e^{-\theta_0}$. Therefore, using the Invariance Property of the MLE, we have that the ML estimator for γ_0 is

$$\widehat{\gamma}_{MLE} = e^{-\widehat{\theta}_{MLE}} = e^{-\overline{X}}$$

To derive the asymptotic distribution of $\widehat{\gamma}_{MLE}$ we use the fact that we know the limiting distribution of $\widehat{\theta}_{MLE}$ (see (5)) and we want to derive the asymptotic distribution of a *function* of $\widehat{\theta}_{MLE}$, which is $\widehat{\gamma}_{MLE} = e^{-\widehat{\theta}_{MLE}}$. Therefore, we use the Delta Method. In this case we have that $\gamma_0 = e^{-\theta_0}$, and therefore

$$\frac{de^{-\theta_0}}{d\theta} = -e^{-\theta_0}$$

Applying the Delta Method we have

$$\sqrt{n}(\widehat{\gamma}_{MLE} - \gamma_0) \xrightarrow{d} N\left(0, \left[-e^{-\theta_0}\right]^2 \theta_0\right),$$

or simply

$$\sqrt{n}(\widehat{\gamma}_{MLE} - \gamma_0) \xrightarrow{d} N\left(0, e^{-2\theta_0}\theta_0\right).$$

Finally, the asymptotic distribution of $\widehat{\gamma}_{MLE}$ is given by

$$\widehat{\gamma}_{MLE} \overset{A}{\sim} N\left(\gamma_0, e^{-2\theta_0}\theta_0/n\right)$$